

DAY 1 — PROJECT SETUP (FROM SCRATCH, WITH REUSE PLAN)

✅ WHERE WE ARE

- New repo ✅
 - New environment ✅
 - Goal: **production-grade engineering intelligence system**
-

✅ WHAT WE COMPLETE TODAY

- repo structure ✅
 - env setup ✅
 - base API ✅
 - plug reusable old code ✅
-

✅ WHAT IS NEXT (DAY 2)

- PDF parsing layer (Unstructured / LlamaParse)
-

✅ FINAL FOLDER STRUCTURE (CREATE THIS EXACT)

project_root/

```
|
|
|─ app/
|
|  └─ api/          ← FastAPI
|    └─ main.py
|
|
|  └─ ingestion/    ← NEW (critical)
|    └─ loader.py
|    └─ pipeline.py
|
|
|  └─ parser/       ← NEW
```

```
| | ├── pdf_parser.py
| | ├── table_parser.py
| | ├── text_parser.py
| |
| | ├── schema/      ← NEW (core)
| |   ├── models.py
| |   └── mapper.py
| |
| | ├── storage/     ← NEW
| |   ├── db.py
| |
| | ├── query/       ← NEW
| |   ├── parser.py
| |   ├── engine.py
| |   └── analytics.py
| |
| | ├── agents/      ← REUSE + UPGRADE
| |   ├── main_agent.py
| |   ├── rag_agent.py
| |   └── table_agent.py
| |
| | ├── models/      ← REUSE
| |   ├── llm.py
| |   └── embeddings.py
| |
| | ├── utils/
|
| ├── data/
```

└─ cache/
└─ logs/
└─ tests/
└─ .env
└─ requirements.txt

✅ WHERE OLD CODE FITS

Old Code	Where to put
FastAPI API	app/api/ ✅
LLM wrapper	app/models/llm.py ✅
Embeddings	app/models/embeddings.py ✅
Existing agent system	app/agents/ ✅
Excel agent	app/agents/ ✅
RAG (basic)	app/agents/rag_agent.py ✅

👉 Based on your template PDF:

✅ you already had **multi-agent structure**

👉 we KEEP it — we don't rewrite [\[w4d23 | PDF\]](#)

✅ ❌ WHAT YOU DO NOT COPY

Remove
PDF chunking logic ❌
embedding-only pipeline ❌
direct RAG-only flow ❌

✅ WHAT YOU ADD TODAY

Create minimal API

app/api/main.py:

```
from fastapi import FastAPI
```

```
app = FastAPI()
```

```
@app.get("/")
```

```
def health():
```

```
    return {"status": "ok"}
```

```
``
```

✓ ✓ ENV SETUP

```
python -m venv venv
```

```
source venv/bin/activate # or windows
```

```
pip install fastapi uvicorn
```

✓ ✓ RUN

```
uvicorn app.api.main:app --reload
```

✓ ✓ VERIFY

Open:

<http://localhost:8000>

✓ ✓ TODAY OUTPUT

✓ Repo ready

✓ Folder structure ready

✓ API running

✓ Old code integrated (core parts only)

✅ 🚀 **STATE AFTER DAY 1**

Infra ✅

Agents ✅ (old reused)

LLM ✅ (old reused)

Parsing ❌ (not yet)

Data layer ❌ (not yet)

Analytics ❌

✅ 🚀 **NEXT (DAY 2)**

👉 Build:

PDF ingestion + parsing base

(Unstructured / LlamaParse)

✅ When ready → say: **Day 2**